

STA 445 S26 Assignment 2

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Load all the packages you will need here. Notice that the code chunk option has been set to include = FALSE. This means the code will run, but the output will not be shown in the knitted document. Thus, you do not need to worry about suppressing warning messages. This code will be helpful when writing a project report.

Problem 1

Download from GitHub the data file by clicking on this link: [Example_5.xls](#). Open it in Excel and figure out which sheet of data we should import into R. At the same time figure out which range of values to import.

- a. Import the dataset into a data frame and show the structure of the imported data using the `str()` command. Make sure that your data has $n = 31$ observations and the three columns are appropriately named. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file.

```
CherryTrees <- read_xls("Example_5.xls", sheet = "RawData", range = "A5:C36")
str(CherryTrees)
```

```
## tibble [31 x 3] (S3: tbl_df/tbl/data.frame)
##  $ Girth : num [1:31] 8.3 8.6 8.8 10.5 10.7 10.8 11 11 11.1 11.2 ...
##  $ Height: num [1:31] 70 65 63 72 81 83 66 75 80 75 ...
##  $ Volume: num [1:31] 10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 ...
```

- b. Use `slice_sample()` to show five random rows of the dataset.

```
slice_sample(CherryTrees, n=5)
```

```
## # A tibble: 5 x 3
##   Girth Height Volume
##   <dbl> <dbl> <dbl>
## 1  20.6     87     77
## 2  12.9     85    33.8
## 3  10.5     72    16.4
## 4   12     75    19.1
## 5  17.9     80    58.3
```

Problem 2

Download from GitHub the data file by clicking on this link: [Example_3.xls](#). Import the data set into a data frame and show the structure of the imported data using the `str()` command. Make sure the Tesla values are NA where appropriate and that both -9999 and NA are imported as NA values. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file.

```
Cars <- read_xls("Example_3.xls", sheet = "data", range = "A1:L34", na=c("-9999", "NA"))
str(Cars)
```

```
## tibble [33 x 12] (S3: tbl_df/tbl/data.frame)
## $ model: chr [1:33] "Mazda RX4" "Mazda RX4 Wag" "Datsun 710" "Hornet 4 Drive" ...
## $ mpg : num [1:33] 21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
## $ cyl : num [1:33] 6 6 4 6 8 6 8 4 4 6 ...
## $ disp : num [1:33] 160 160 108 258 360 ...
## $ hp : num [1:33] 110 110 93 110 175 105 245 62 95 123 ...
## $ drat : num [1:33] 3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
## $ wt : num [1:33] 2.62 2.88 2.32 3.21 3.44 ...
## $ qsec : num [1:33] 16.5 17 18.6 19.4 17 ...
## $ vs : num [1:33] 0 0 1 1 0 1 0 1 1 1 ...
## $ am : num [1:33] 1 1 1 0 0 0 0 0 0 0 ...
## $ gear : num [1:33] 4 4 4 3 3 3 3 4 4 4 ...
## $ carb : num [1:33] 4 4 1 1 2 1 4 2 2 4 ...
```

b. Use the `tail()` command to show the last five rows of a data table.

```
tail(Cars, n=5)
```

```
## # A tibble: 5 x 12
##   model      mpg   cyl  disp    hp  drat    wt   qsec    vs  am  gear  carb
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Ford Panter~ 15.8     8   351   264  4.22  3.17  14.5     0    1    5     4
## 2 Ferrari Dino 19.7     6   145   175  3.62  2.77  15.5     0    1    5     6
## 3 Maserati Bo~ 15      8   301   335  3.54  3.57  14.6     0    1    5     8
## 4 Volvo 142E  21.4     4   121   109  4.11  2.78  18.6     1    1    4     2
## 5 Tesla Model~ 98      NA    NA   778  NA    4.94  10.4    NA    0    1    NA
```

c. Use `slice_sample()` to show five random rows of the dataset.

```
slice_sample(Cars, n=5)
```

```
## # A tibble: 5 x 12
##   model      mpg   cyl  disp    hp  drat    wt   qsec    vs  am  gear  carb
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Dodge Chall~ 15.5     8  318    150  2.76  3.52  16.9     0    0    3     2
## 2 Fiat 128     32.4     4  78.7     66  4.08  2.2   19.5     1    1    4     1
## 3 Maserati Bo~ 15      8  301    335  3.54  3.57  14.6     0    1    5     8
## 4 Porsche 914~ 26      4  120.     91  4.43  2.14  16.7     0    1    5     2
## 5 Ford Panter~ 15.8     8  351    264  4.22  3.17  14.5     0    1    5     4
```

Problem 3

Download all of the files from GitHub `data-raw/InsectSurveys` directory which have been linked here. Each month's file contains a sheet with information about each of the sites that was surveyed. The second sheet contains information about the number of each species that was observed at each site.

- Import the data for each month and create a single `site` data frame with information from each month. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file. You may need to change column names and reorganize columns.

```
May <- read_xlsx("May.xlsx", sheet = "Sites", range = "A1:F10", .name_repair = toupper)
June <- read_xlsx("June.xlsx", sheet = "Sites", range = "A1:F10", .name_repair = toupper)
## change to Water_Depth and PH
July <- read_xlsx("July.xlsx", sheet = "Sites", range = "A1:F10", .name_repair = toupper)
August <- read_xlsx("August.xlsx", sheet = "sites", range = "A1:F10", .name_repair = toupper)
September <- read_xlsx("September.xlsx", sheet = "sites", range = "A1:F10", .name_repair = toupper)
## date isn't in right format
October <- read_xlsx("October.xlsx", sheet = "Sites", range = "A1:F10", .name_repair = toupper, n

July <- July %>% select("SITE NAME", "POND AREA", "WATER DEPTH", "PH", "DATE", "OBSERVER")

site <- rbind(May, June, July, August, September, October)
```

- Show the structure of the data frame using the `str()` command.

```
str(site)

## tibble [54 x 6] (S3: tbl_df/tbl/data.frame)
##  $ SITE NAME   : chr [1:54] "Araphahoe Road" "Bridger Valley" "Calculus Vector" "Deer Valley" .
##  $ POND AREA   : num [1:54] 34 240 321 74 28 62 489 398 126 34 ...
##  $ WATER DEPTH: num [1:54] 3 6 13 4.4 2 3.6 4 10 9 3 ...
##  $ PH          : num [1:54] 6.2 6.5 6.4 6.9 7.1 7 7.1 6.8 6.75 6.2 ...
##  $ DATE        : POSIXct[1:54], format: "2020-05-15" "2020-05-16" ...
##  $ OBSERVER    : chr [1:54] "Bob" "Bob" "Bob" "Bob" ...
```

- Use `slice_sample()` to show eight random rows of the dataset.

```
slice_sample(site, n=8)

## # A tibble: 8 x 6
##   'SITE NAME'      'POND AREA' 'WATER DEPTH'    PH DATE                OBSERVER
##   <chr>            <dbl>         <dbl> <dbl> <dtm>                 <chr>
## 1 Fennel Gardens    62            3.6  7    2020-07-16 00:00:00 Charlie
## 2 Araphahoe Road    34            3    6.2  2020-08-15 00:00:00 Bob
## 3 Fennel Gardens    62            3.6  7    2020-08-16 00:00:00 Charlie
## 4 Fennel Gardens    62            3.6  7    NA                      Charlie
```

## 5 Happy Feet	398	10	6.8	2020-06-18 00:00:00	Charlie
## 6 Indigo Flats	126	9	6.75	2020-05-19 00:00:00	Charlie
## 7 Calculus Vector	321	13	6.4	2020-07-17 00:00:00	Bob
## 8 Bridger Valley	240	6	6.5	2020-09-16 00:00:00	Bob

- d. Comment on the importance of consistency of your data input sheets. Due to the input sheets being consistent, it allowed for easier manipulation and modification to the data set.